# Stochastic Calculus & Option Pricing — Synthetic Notes

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**How to use these notes.**

- These are *synthetic* lecture notes: definitions, key results, and the computations you actually use.
- Each lecture ends with a short checklist: "what you must be able to do".
- Notation: all processes are defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions.

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# 1 Lecture 1 — Probability refresher and core mathematical tools

**Summary.** Random variables and their laws; CDFs and densities; moments, variance, and covariance; dominated convergence; LLN and CLT; characteristic functions; information and conditional expectation; Gaussian conditioning; filtrations and martingales; and the differential-calculus formulas used later.

**Reference.** S. E. Shreve, *Stochastic Calculus for Finance II*, Chapters 1–2.

## 1.1 Random variables, laws, CDFs, and densities

**Random variable and law.** On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a real random variable is a measurable map $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its law is the probability measure

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \qquad A \in \mathcal{B}(\mathbb{R}).$$

The random variable maps states to numbers; its law records the probabilities of those numbers.

**CDF and density.** The cumulative distribution function of $X$ is

$$F_X(x) = \mathbb{P}(X \le x).$$

Every CDF is nondecreasing and right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$. For $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

If $X$ has density $f_X$, then

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,\mathrm{d}u, \qquad \mathbb{P}(a < X \le b) = \int_a^b f_X(u)\,\mathrm{d}u,$$

and $f_X = F_X'$ wherever the CDF is differentiable. A law and a CDF always exist; a density need not exist.

**Joint and marginal laws.** The joint law of $(X, Y)$ contains both marginal laws and their dependence. In the discrete case,

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y),$$

whereas for a joint density,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x.$$

Independence is equivalent to factorization of the joint law: $p_{X,Y} = p_X p_Y$ or $f_{X,Y} = f_X f_Y$.

**Common one-dimensional laws.**

| Law | Parameters | $\mathbb{E}[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|
| Bernoulli | $p$ | $p$ | $p(1-p)$ |
| Binomial | $n, p$ | $np$ | $np(1-p)$ |
| Poisson | $\lambda$ | $\lambda$ | $\lambda$ |
| Uniform | $[a, b]$ | $(a+b)/2$ | $(b-a)^2/12$ |
| Exponential | rate $\lambda$ | $1/\lambda$ | $1/\lambda^2$ |
| Gaussian | $\mu, \sigma^2$ | $\mu$ | $\sigma^2$ |

**Mass functions and densities to recognize.** For the discrete laws,

$$\begin{aligned}
X \sim \mathrm{Bernoulli}(p): &\quad \mathbb{P}(X = x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\},\\
X \sim \mathrm{Binomial}(n, p): &\quad \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n,\\
X \sim \mathrm{Poisson}(\lambda): &\quad \mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k \in \mathbb{N}.
\end{aligned}$$

For the continuous laws,

$$\begin{aligned}
X \sim \mathrm{Unif}[a, b]: &\quad f_X(x) = \frac{1}{b-a}\mathbf{1}_{[a,b]}(x),\\
X \sim \mathrm{Exp}(\lambda): &\quad f_X(x) = \lambda e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x),\\
X \sim \mathcal{N}(\mu, \sigma^2): &\quad f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).
\end{aligned}$$

## 1.2 Expectation, moments, variance, and covariance

**Expectation of a transformation.** If the relevant sum or integral is finite, then

$$\mathbb{E}[h(X)] = \sum_x h(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x) f_X(x)\,\mathrm{d}x.$$

The $k$th moment is $\mathbb{E}[X^k]$. Integrability means $\mathbb{E}[|X|] < \infty$, and $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

**Essential expectation properties.** For integrable $X, Y$ and constants $a, b$,

$$\begin{aligned}\mathbb{E}[aX + bY] &= a\mathbb{E}[X] + b\mathbb{E}[Y],\\ X \le Y &\Longrightarrow \mathbb{E}[X] \le \mathbb{E}[Y],\\ |\mathbb{E}[X]| &\le \mathbb{E}[|X|].\end{aligned}$$

Linearity does not require independence.

**Variance and covariance.** For square-integrable variables,

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)},$$

and

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The identities used most often are

$$\begin{aligned}\mathrm{Var}(aX + b) &= a^2\mathrm{Var}(X),\\ \mathrm{Var}(X + Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y),\\ \mathrm{Var}\left(\sum_{k=1}^{n} X_k\right) &= \sum_{k=1}^{n} \mathrm{Var}(X_k) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j).\end{aligned}$$

Independence implies zero covariance, but the converse is false in general. For independent square-integrable variables, the variance of a sum is the sum of the variances.

**Theorem 1.1** (Dominated Convergence Theorem)**.** *Suppose $X_n \to X$ almost surely and there is an integrable random variable $Y$ such that $|X_n| \le Y$ almost surely for every $n$. Then $X$ is integrable and*

$$\mathbb{E}[|X_n - X|] \longrightarrow 0, \qquad \mathbb{E}[X_n] \longrightarrow \mathbb{E}[X].$$

*Remark* 1.2. Dominated convergence is the standard tool for interchanging a limit and an expectation. Pointwise convergence alone is not sufficient; the integrable dominating variable controls the tails uniformly in $n$.

## 1.3 Sample moments, LLN, and CLT

For observations $X_1, \ldots, X_n$, define

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad v_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2, \qquad s_n^2 = \frac{n}{n-1}v_n.$$

Here $v_n$ is the empirical variance and $s_n^2$ is its usual unbiased version under i.i.d. sampling with finite variance.

**Theorem 1.3** (Law of Large Numbers)**.** *If $(X_i)$ are i.i.d. and $\mathbb{E}[|X_1|] < \infty$, then*

$$\overline{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu = \mathbb{E}[X_1].$$

**Theorem 1.4** (Central Limit Theorem)**.** *If, in addition, $0 < \sigma^2 = \mathrm{Var}(X_1) < \infty$, then*

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0, 1).$$

*Thus $\overline{X}_n$ has approximate standard deviation $\sigma/\sqrt{n}$ for large $n$.*

The LLN says that the estimation error vanishes; the CLT gives its $n^{-1/2}$ scale and approximate Gaussian shape.

## 1.4 Characteristic functions

**Definition and basic properties.** The characteristic function of $X$ is

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \qquad u \in \mathbb{R}.$$

It always exists because $|e^{iuX}| = 1$. Moreover,

$$\varphi_X(0) = 1, \qquad |\varphi_X(u)| \le 1, \qquad \varphi_{aX+b}(u) = e^{iub}\varphi_X(au).$$

The characteristic function uniquely determines the law. If $\mathbb{E}[|X|^k] < \infty$, then it can be differentiated at the origin and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k].$$

**Independent sums.** If $X$ and $Y$ are independent, then

$$\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u).$$

This turns convolution of laws into multiplication of functions.

**Formulas to know.**

$$\begin{aligned}
X \sim \text{Bernoulli}(p): &\quad \varphi_X(u) = 1 - p + pe^{iu}, \\
X \sim \text{Poisson}(\lambda): &\quad \varphi_X(u) = \exp\big(\lambda(e^{iu} - 1)\big), \\
X \sim \text{Exp}(\lambda): &\quad \varphi_X(u) = \frac{\lambda}{\lambda - iu}, \\
X \sim \mathcal{N}(\mu, \sigma^2): &\quad \varphi_X(u) = \exp\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right).
\end{aligned}$$

For $Z \sim \mathcal{N}(0,1)$, differentiation under the integral followed by integration by parts gives

$$\varphi_Z'(u) = -u\varphi_Z(u), \qquad \varphi_Z(0) = 1.$$

Hence $\varphi_Z(u) = e^{-u^2/2}$; writing $X = \mu + \sigma Z$ gives the general Gaussian formula.

## 1.5 Information, measurability, and conditional expectation

**$\sigma$-algebras as information.** A sub-$\sigma$-algebra $\mathcal{G} \subseteq \mathcal{F}$ represents the events that can be decided from the available information. A variable $Z$ is $\mathcal{G}$-measurable when its value can be determined from that information. The information revealed by a signal $Y$ is $\sigma(Y)$.

On a finite space, $\mathcal{G}$ is generated by a partition $A_1, \dots, A_m$. A variable is $\mathcal{G}$-measurable if and only if it is constant on every atom $A_j$. Equivalently,

$$Z \text{ is } \sigma(Y)\text{-measurable} \quad \Longleftrightarrow \quad Z = g(Y)$$

for some measurable $g$.

**Conditional probability and conditional laws.** For events $A, B$ with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For discrete variables and $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

For a joint density and $f_Y(y) > 0$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x.$$

**Definition of conditional expectation.** For $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the almost-surely unique random variable $Z$ such that

$$Z \text{ is } \mathcal{G}\text{-measurable}, \qquad \int_A Z\,\mathrm{d}\mathbb{P} = \int_A X\,\mathrm{d}\mathbb{P} \quad \text{for every } A \in \mathcal{G}.$$

It is the prediction of $X$ based only on the information in $\mathcal{G}$. Conditioning on $Y$ means conditioning on $\sigma(Y)$:

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

**Finite-partition formula.** If $\mathcal{G} = \sigma(A_1, \ldots, A_m)$, where the $A_j$ are partition atoms with positive probability, then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{j=1}^{m} \frac{\mathbb{E}[X\mathbf{1}_{A_j}]}{\mathbb{P}(A_j)}\mathbf{1}_{A_j} = \sum_{j=1}^{m} \mathbb{E}[X \mid A_j]\mathbf{1}_{A_j}.$$

Conditional expectation is therefore the average of $X$ inside each information set.

**Formula for $\mathbb{E}[h(X) \mid Y]$: discrete case.** Define, for every $y$ with $\mathbb{P}(Y = y) > 0$,

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) \mid Y = y] \\ &= \sum_x h(x)\mathbb{P}(X = x \mid Y = y) \\ &= \frac{\sum_x h(x)\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}. \end{aligned}$$

Then

$$\boxed{\mathbb{E}[h(X) \mid Y] = m(Y).}$$

**Formula for $\mathbb{E}[h(X) \mid Y]$: density case.** For $f_Y(y) > 0$, define

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) \mid Y = y] \\ &= \int_{\mathbb{R}} h(x) f_{X|Y}(x \mid y)\,\mathrm{d}x \\ &= \frac{\int_{\mathbb{R}} h(x) f_{X,Y}(x, y)\,\mathrm{d}x}{\int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x}. \end{aligned}$$

Again,

$$\boxed{\mathbb{E}[h(X) \mid Y] = m(Y).}$$

The final substitution $y = Y$ is essential: a conditional expectation is a random variable, not merely a number indexed by $y$.

**Essential conditional-expectation properties.** For integrable variables, constants $a, b$, and $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$,

$$\begin{aligned}
\mathbb{E}[aX + bY \mid \mathcal{G}] &= a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}], \\
X \le Y &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}], \\
Z \ \mathcal{G}\text{-measurable} &\Longrightarrow \mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}], \\
X \ \mathcal{G}\text{-measurable} &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] = X, \\
X \text{ independent of } \mathcal{G} &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X], \\
\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] &= \mathbb{E}[X \mid \mathcal{H}].
\end{aligned}$$

In particular,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

For convex $\phi$, conditional Jensen gives

$$\phi(\mathbb{E}[X \mid \mathcal{G}]) \le \mathbb{E}[\phi(X) \mid \mathcal{G}]$$

whenever the quantities are integrable.

**Gaussian conditioning.** If $(X, Y)$ is jointly Gaussian, with standard deviations $\sigma_X, \sigma_Y > 0$ and correlation $\rho$, then

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1 - \rho^2)\right).$$

Consequently,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \qquad \mathrm{Var}(X \mid Y) = \sigma_X^2(1 - \rho^2).$$

The conditional mean is affine in the observed signal, and the residual variance decreases as $|\rho|$ increases.

### 1.6 Processes, filtrations, and martingales

**Processes and filtrations.** A discrete-time stochastic process is a sequence $(X_n)_{n \ge 0}$ of random variables on the same probability space. A filtration is an increasing family of $\sigma$-algebras,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots,$$

representing information accumulated through time. The natural filtration of $X$ is

$$\mathcal{F}_n^X = \sigma(X_0, \ldots, X_n).$$

A process is adapted if $X_n$ is $\mathcal{F}_n$-measurable for every $n$. Adaptedness is the no-look-ahead requirement.

**Random walks.** If

$$S_n = S_0 + \sum_{k=1}^{n} \xi_k,$$

where the increments are i.i.d. with mean $m$ and variance $\sigma^2$, then

$$\mathbb{E}[S_n] = S_0 + nm, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

**Martingales.** An integrable adapted process $(M_n)$ is a martingale if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for every } n.$$

For the random walk above,

$$M_n = S_n - nm$$

is a martingale. For independent symmetric $\pm 1$ increments, $S_n^2 - n$ is also a martingale. A martingale combines three requirements: integrability, adaptation, and the conditional fair-game identity.

## 1.7 Differential-calculus review

**One-variable derivative and Taylor expansion.** For $g : \mathbb{R} \to \mathbb{R}$,

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}.$$

If $g$ is twice differentiable near $x$, then

$$g(x+h) = g(x) + g'(x)h + \frac{1}{2} g''(x)h^2 + o(h^2).$$

**Gradient and Hessian.** For $f : \mathbb{R}^d \to \mathbb{R}$,

$$\nabla f(x) = \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_d f(x) \end{pmatrix}, \qquad \nabla^2 f(x) = \big(\partial_{ij} f(x)\big)_{1 \le i,j \le d}.$$

The gradient gives first-order sensitivities; the Hessian records curvature and interactions. The second-order multivariable expansion is

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + o(\|h\|^2).$$

**Multivariable chain rule.** If $x(t) \in \mathbb{R}^d$ and $f$ are differentiable, then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t)) = \nabla f(x(t))^\top x'(t).$$

In particular, for $V = V(t,s)$ and a differentiable path $S(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} V(t, S(t)) = V_t(t, S(t)) + V_s(t, S(t)) S'(t).$$

The second-order Taylor term involving $V_{ss}$ will be essential once the underlying path is Brownian rather than differentiable.

**Lecture 1 checklist**

You should be able to:

- pass between a law, a CDF, a density, and interval probabilities;
- compute $\mathbb{E}[h(X)]$, moments, variances, covariances, and variances of sums;
- state when dominated convergence allows a limit to pass through $\mathbb{E}$;
- distinguish the conclusions of the LLN and the CLT;
- compute and use characteristic functions, especially for independent sums;
- identify the information in $\sigma(Y)$ and test measurability on partition atoms;
- compute $\mathbb{E}[h(X) \mid Y]$ in both discrete and density models;
- use linearity, known factors, independence, monotonicity, Jensen, and the tower rule;
- apply the Gaussian conditioning formula;
- verify adaptation and the martingale property;
- compute derivatives, gradients, Hessians, Taylor expansions, and chain rules.

# 2 Lecture 2 — Brownian Motion, Quadratic Variation, and GBM

**References**

- *Stochastic Calculus for Finance II: Continuous-Time Models*, S. E. Shreve, Chapter 3, pp. 83-117
- *Brownian Motion and Stochastic Calculus*, I. Karatzas and S. E. Shreve, Chapter 2, pp. 47-126

## 2.1 From Discrete Random Walks to Brownian Motion

**Simple symmetric random walk.** Let $(\varepsilon_k)_{k\geq 1}$ be i.i.d. with $\mathbb{P}(\varepsilon_k = \pm 1) = \frac{1}{2}$, and define

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} \varepsilon_k.$$

Then $(S_n)$ is a discrete-time martingale with independent increments, $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$.

**Diffusive scaling.** Introduce a time step $1/n$ and define the rescaled process

$$W^{(n)}(t) = \frac{1}{\sqrt{n}}\, S_{\lfloor nt \rfloor}, \qquad t \geq 0,$$

with linear interpolation. By the CLT, increments of $W^{(n)}$ over intervals of length $t - s$ are approximately Gaussian with variance $t - s$.

**Donsker invariance principle (informal).** As $n \to \infty$, $W^{(n)}$ converges in distribution to a continuous-time process $W$, called *Brownian motion*. This result explains why Brownian motion is the universal continuous-time limit of random walks.

## 2.2 Definition and Basic Properties of Brownian Motion

**Definition.** A process $(W_t)_{t\geq 0}$ is a *standard Brownian motion* if:

- $W_0 = 0$,
- it has independent increments,
- for $0 \leq s < t$, $W_t - W_s \sim \mathcal{N}(0, t-s)$,
- paths are continuous almost surely.

**Gaussian process viewpoint.** Brownian motion is a centered Gaussian process fully characterized by

$$\mathrm{Cov}(W_s, W_t) = \min(s,t).$$

In particular, for $s < t$,

$$W_t = W_s + (W_t - W_s),$$

where $W_s$ and $W_t - W_s$ are independent.

**Martingale property.** With respect to its natural filtration $\mathcal{F}_t = \sigma(W_s : s \leq t)$, Brownian motion is a martingale:

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s, \qquad s \leq t.$$

## 2.3 Markov and Strong Markov Properties

**Markov property.** Brownian motion is Markov: conditionally on $W_s$, the future is independent of the past, if $t \geq s$,

$$\mathbb{P}(W_t \in A \mid \mathcal{F}_s) = \mathbb{P}(W_t \in A \mid W_s).$$

Equivalently,

$$W_t \mid W_s = x \sim \mathcal{N}(x, t-s).$$

**Strong Markov property.** Let $\tau$ be a stopping time. Conditionally on $\mathcal{F}_\tau$, the shifted process

$$(W_{\tau+t} - W_\tau)_{t\geq 0}$$

is a Brownian motion independent of $\mathcal{F}_\tau$. This is crucial for hitting times, barriers, and American-style features.

## 2.4 Reflection Principle and Hitting Times

**Running maximum.** Define the running maximum

$$M_t = \sup_{0\leq s\leq t} W_s,$$

and the hitting time of level $a$:

$$\tau_a = \inf\{t \geq 0 : \ W_t = a\}.$$

**Reflection principle.** For $a > 0$ and $t > 0$,

$$\mathbb{P}(M_t \geq a) = 2\,\mathbb{P}(W_t \geq a).$$

More generally, for $w \leq a$,

$$\mathbb{P}(M_t > a,\ W_t < w) = \mathbb{P}(W_t > 2a - w).$$

**Interpretation.** Among paths that hit level $a$ before time $t$, reflecting the path after the first hitting time preserves Wiener measure. This converts path-dependent events into Gaussian tail probabilities.

**Consequence.** Brownian motion hits every level almost surely:

$$\mathbb{P}(\tau_a < \infty) = 1, \qquad a \in \mathbb{R}.$$

### 2.5 Pathwise Regularity and Irregularity

**Nowhere differentiability.** Almost surely, Brownian paths are nowhere differentiable.

**Oscillations.** For every $t \geq 0$, almost surely,

$$\limsup_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = +\infty, \qquad \liminf_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = -\infty.$$

**Hölder regularity.** For every $\alpha < \frac{1}{2}$, Brownian motion is locally $\alpha$-Hölder, but it is not $\frac{1}{2}$-Hölder.

### 2.6 Quadratic Variation

**Definition.** For a partition $\pi = \{0 = t_0 < \cdots < t_n = T\}$, define

$$[X]_T^\pi = \sum_{k=0}^{n-1} (X_{t_{k+1}} - X_{t_k})^2.$$

If this converges as $|\pi| \to 0$, the limit is the quadratic variation $[X]_T$.

**Finite variation vs. Brownian motion.** If $X_t = \int_0^t v_s ds$, then $[X]_T = 0$. In contrast, for Brownian motion,

$$[W]_T = T.$$

**Key martingale.** Quadratic variation explains why

$$W_t^2 - t$$

is a martingale. This is the simplest manifestation of Itô calculus.

**Quadratic covariation.** For processes $X, Y$,

$$[X, Y]_T = \lim_{\pi} \sum_k (X_{t_{k+1}} - X_{t_k})(Y_{t_{k+1}} - Y_{t_k}).$$

If $A$ has finite variation, then $[A, W] = 0$.

## 2.7 Heuristic Itô Rules

Brownian increments satisfy the scaling:

$$\Delta W = O(\sqrt{\Delta t}).$$

As a bookkeeping device,

$$(dW_t)^2 = dt, \qquad dW_t\, dt = 0, \qquad (dt)^2 = 0.$$

These rules encode which discrete sums survive in the limit.

## 2.8 Geometric Brownian Motion (GBM)

**Motivation.** Additive models allow negative prices and ignore scale effects. Instead, model relative changes:

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t.$$

**Explicit solution.** The solution is

$$S_t = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\Big),$$

so $S_t$ is lognormally distributed.

**Moments.**

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \mathrm{Var}(S_t) = S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1).$$

**Volatility scaling.** Log-returns over $\Delta t$ have variance $\sigma^2 \Delta t$, which justifies the $\sqrt{\Delta t}$ rule.

## 2.9 Stopping Times and Optional Stopping

**Stopped process.** For a stopping time $\tau$ and a martingale $(M_t)$, define

$$M_t^\tau = M_{t \wedge \tau}.$$

Then $(M_t^\tau)$ is again a martingale.

**Optional stopping (safe version).** If $\tau$ is bounded, then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

In general, $\tau < \infty$ a.s. is not sufficient; integrability or uniform integrability is required.

**Example.** For Brownian motion and $\tau_a = \inf\{t : \ W_t = a\}$, $\mathbb{E}[\tau_a] = \infty$ and $W_{\tau_a} = a$, so $\mathbb{E}[W_{\tau_a}] \neq \mathbb{E}[W_0]$. Optional stopping applies only to $t \wedge \tau_a$.

## 2.10 Multidimensional Brownian Motion

**Definition.** A $d$-dimensional Brownian motion is a vector-valued process

$$W_t = (W_t^1, \dots, W_t^d), \qquad t \geq 0,$$

such that:

- $W_0 = 0 \in \mathbb{R}^d$,
- it has independent increments,
- for $0 \leq s < t$, the increment $W_t - W_s$ is multivariate Gaussian with mean 0 and covariance matrix $(t-s)I_d$,
- paths are continuous almost surely.

Equivalently, the components $(W^1, \dots, W^d)$ are independent one-dimensional Brownian motions.

**Gaussian characterization.** Multidimensional Brownian motion is a centered Gaussian process with

$$\operatorname{Cov}(W_s^i, W_t^j) = \delta_{ij} \min(s,t), \qquad 1 \leq i, j \leq d,$$

where $\delta_{ij}$ is the Kronecker symbol.

**Independent vs. correlated Brownian motions.** More generally, one may consider a Brownian motion with covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric, positive semidefinite), defined by

$$\operatorname{Cov}(W_s, W_t) = \min(s,t)\, \Sigma.$$

In this case,

$$\mathrm{d}[W^i, W^j]_t = \Sigma_{ij}\, \mathrm{d}t.$$

Any such process can be written as

$$W_t = B\, \widetilde{W}_t,$$

where $\widetilde{W}$ is a standard $d$-dimensional Brownian motion and $B$ satisfies $BB^\top = \Sigma$ (e.g. Cholesky factorization).

**Quadratic variation and covariation.** For a $d$-dimensional Brownian motion $W$,

$$[W^i]_t = t, \qquad [W^i, W^j]_t = 0 \quad (i \neq j)$$

in the independent case, and more generally

$$[W]_t = t\, \Sigma,$$

where $[W]_t$ denotes the matrix-valued quadratic variation.

**Itô rules (multidimensional mnemonic).** For independent components,

$$(\mathrm{d}W_t^i)^2 = \mathrm{d}t, \qquad \mathrm{d}W_t^i\, \mathrm{d}W_t^j = 0 \ (i \neq j), \qquad \mathrm{d}W_t^i\, \mathrm{d}t = 0, \qquad (\mathrm{d}t)^2 = 0.$$

With correlation matrix $\Sigma$, this becomes

$$\mathrm{d}W_t^i\, \mathrm{d}W_t^j = \Sigma_{ij}\, \mathrm{d}t.$$

**Preview.** In higher dimension, stochastic calculus keeps the same structure: *drift terms*, *martingale terms*, and *quadratic variation*. Only the bookkeeping becomes matrix-valued. This naturally leads to the multidimensional Itô formula and to multidimensional diffusion models.

### Lecture 2 checklist

You should be able to:

- state the defining properties of Brownian motion and compute its Gaussian increment laws and covariance function;
- use independent increments to verify the martingale and Markov properties;
- apply the reflection principle to convert a maximum or hitting-time event into a Gaussian tail probability;
- distinguish Brownian paths from finite-variation paths, in particular through their quadratic variation;
- use the Itô multiplication rules $(\mathrm{d}W_t)^2 = \mathrm{d}t$, $\mathrm{d}W_t\,\mathrm{d}t = 0$, and $(\mathrm{d}t)^2 = 0$;
- solve the geometric Brownian-motion SDE and identify the lognormal law of $S_t$;
- identify a stopping time and state a safe version of optional stopping, including its hypotheses;
- work with independent and correlated multidimensional Brownian motions and their covariations.

# 3 Lecture 3 — The Itô Stochastic Integral

**Summary.** The Itô integral is the $L^2$ limit of left-point, non-anticipative gains. Predictability prevents the use of future information and finite expected energy gives integrability. The resulting gain process is a square-integrable martingale, with quadratic variation $\int_0^t \phi_s^2\,\mathrm{d}s$; for deterministic integrands, $L^2$ inner products become covariances of jointly Gaussian variables.

## 3.1 Definition, admissibility, and construction

**Why left endpoints are essential.** For a grid $0 = t_0 < \cdots < t_n = T$, a discrete gain has the form

$$G_T^\pi = \sum_{i=0}^{n-1} \phi_{t_i}(W_{t_{i+1}} - W_{t_i}).$$

The coefficient used on $(t_i, t_{i+1}]$ must be known at time $t_i$: $\phi_{t_i}$ is $\mathcal{F}_{t_i}$-measurable. This is the mathematical content of predictability and the economic content of non-anticipation. The sampling rule matters because Brownian quadratic variation survives:

$$\sum_i W_{t_{i+1}}\Delta W_i - \sum_i W_{t_i}\Delta W_i = \sum_i (\Delta W_i)^2 \longrightarrow T.$$

**Definition 3.1** (Simple predictable integrand)**.** A simple predictable process on $[0, T]$ is

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t), \qquad \xi_i \in \mathcal{F}_{t_i}, \qquad \mathbb{E}[\xi_i^2] < \infty.$$

Its Itô integral is the left-point gain

$$\boxed{\int_0^T \phi_t \, \mathrm{d}W_t = \sum_{i=0}^{n-1} \xi_i (W_{t_{i+1}} - W_{t_i}).}$$

**Admissible class and extension.** The standard $L^2$ theory assumes

$$\phi \text{ predictable}, \qquad \boxed{\mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t < \infty.}$$

Choose simple predictable $\phi^n$ such that $\mathbb{E} \int_0^T |\phi_t^n - \phi_t|^2 \, \mathrm{d}t \to 0$. Then

$$\boxed{\int_0^T \phi_t \, \mathrm{d}W_t := L^2\text{-}\lim_{n \to \infty} \int_0^T \phi_t^n \, \mathrm{d}W_t.}$$

The Itô isometry below proves that the limit exists, is unique, and does not depend on the approximating sequence.

### 3.2 The identities that do the calculations

Let $\phi, \psi$ be predictable with finite expected energy. Then

$$\begin{aligned}
\int_0^T (a\phi_t + b\psi_t) \, \mathrm{d}W_t &= a \int_0^T \phi_t \, \mathrm{d}W_t + b \int_0^T \psi_t \, \mathrm{d}W_t, \\
\mathbb{E}\left[ \int_0^T \phi_t \, \mathrm{d}W_t \right] &= 0, \\
\mathbb{E}\left[ \left( \int_0^T \phi_t \, \mathrm{d}W_t \right)^2 \right] &= \mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t, \\
\mathbb{E}\left[ \left( \int_0^T \phi_t \, \mathrm{d}W_t \right) \left( \int_0^T \psi_t \, \mathrm{d}W_t \right) \right] &= \mathbb{E} \int_0^T \phi_t \psi_t \, \mathrm{d}t.
\end{aligned}$$

The third line is the *Itô isometry*; the fourth is its polarization or covariance identity. Thus integrand energy becomes terminal variance.

**Theorem 3.2** (Martingale gain process)**.** *If $\phi$ is predictable and $\mathbb{E} \int_0^T \phi_s^2 \, \mathrm{d}s < \infty$, then*

$$M_t = \int_0^t \phi_s \, \mathrm{d}W_s$$

*is a continuous square-integrable martingale. In particular, for $s \le t$,*

$$\boxed{\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad \mathbb{E}\left[ \int_s^t \phi_u \, \mathrm{d}W_u \middle| \mathcal{F}_s \right] = 0.}$$

The quadratic variation and mutual covariation are

$$\boxed{[M]_t = \int_0^t \phi_s^2 \, \mathrm{d}s, \qquad \left[ \int_0^\cdot \phi_s \, \mathrm{d}W_s, \int_0^\cdot \psi_s \, \mathrm{d}W_s \right]_t = \int_0^t \phi_s \psi_s \, \mathrm{d}s.}$$

The first formula is pathwise; the isometry is its expected-value analogue.

*Remark* 3.3 (True versus local martingale). The expected condition $\mathbb{E} \int_0^T \phi_s^2 \, \mathrm{d}s < \infty$ gives a true $L^2$-martingale. Pathwise local square integrability generally gives only a local martingale; zero drift alone does not justify erasing the stochastic integral after taking expectations.

### 3.3 Deterministic integrands and formulas to know

If $f \in L^2(0,T)$ is deterministic, then

$$\boxed{\int_0^T f(t)\,\mathrm{d}W_t \sim \mathcal{N}\left(0, \int_0^T f(t)^2\,\mathrm{d}t\right).}$$

For deterministic $f, g$, the two integrals are jointly Gaussian and

$$\mathrm{Cov}\left(\int_0^T f\,\mathrm{d}W, \int_0^T g\,\mathrm{d}W\right) = \int_0^T f(t)g(t)\,\mathrm{d}t.$$

Consequently, orthogonality in $L^2(0,T)$ implies independence of the two Gaussian integrals. This implication is special to deterministic integrands; a random predictable integrand need not produce a Gaussian integral.

Three reusable identities are

$$\boxed{\int_0^T c\,\mathrm{d}W_t = cW_T, \qquad \int_0^T t\,\mathrm{d}W_t \sim \mathcal{N}\left(0, \frac{T^3}{3}\right), \qquad \int_0^T W_t\,\mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).}$$

For deterministic absolutely continuous $f$, stochastic integration by parts reduces to

$$\boxed{\int_0^T f(t)\,\mathrm{d}W_t = f(T)W_T - \int_0^T W_t f'(t)\,\mathrm{d}t.}$$

There is no covariation correction because a deterministic finite-variation function has zero covariation with Brownian motion.

If $\tau$ is a stopping time, then on every bounded horizon

$$\boxed{\int_0^t \mathbf{1}_{\{s\le\tau\}}\,\mathrm{d}W_s = W_{t\wedge\tau}, \qquad \mathbb{E}[W_{t\wedge\tau}^2] = \mathbb{E}[t\wedge\tau].}$$

**Reusable calculation pattern.** Before computing an integral, check in order:

1. predictability: is the coefficient known before the increment?
2. finite expected energy: is $\mathbb{E}\int_0^T \phi_t^2\,\mathrm{d}t < \infty$?
3. structure: is the integrand deterministic, stopped, or simple?
4. target: use the isometry for second moments, covariance for joint risks, and the martingale property for conditional means.

### Lecture 3 checklist

You should be able to:

- define an Itô integral first for a simple predictable process;
- check predictability and the condition $\mathbb{E}\int_0^T \phi_t^2\,\mathrm{d}t < \infty$;
- use linearity, zero mean, the Itô isometry, and the covariance identity;

- explain why $\int_0^t \phi_s \, \mathrm{d}W_s$ is a martingale;
- identify the normal law of an integral with deterministic integrand;
- compute brackets of Itô integrals;
- use deterministic integration by parts and derive $\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T)$;
- simulate an Itô integral with left-point Brownian increments and validate it with the isometry.

# 4 Lecture 4 — Itô Processes and Itô's Formula

**Summary.** Quadratic variation makes stochastic calculus second order, and covariation turns correlation into a $\mathrm{d}t$ term. Itô's formula is a chain rule plus a curvature correction. Its product and ratio forms solve SDEs, support discounting and delta hedging, and distinguish local from true martingales.

**Differential versus integral form.** For an Itô process, the compact notation and its precise meaning are

$$\boxed{\mathrm{d}X_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}W_t \quad \Longleftrightarrow \quad X_t - X_s = \int_s^t a_u \, \mathrm{d}u + \int_s^t b_u \, \mathrm{d}W_u, \qquad 0 \le s \le t.}$$

Use the differential form for calculations and the integral form as the rigorous equality. Never divide through by $\mathrm{d}t$: Brownian paths are not differentiable.

## 4.1 Quadratic covariation and the differential table

For continuous processes $X, Y$, quadratic covariation is the limiting sum

$$[X, Y]_t = \lim_{|\pi| \to 0} \sum_i (X_{t_{i+1}} - X_{t_i})(Y_{t_{i+1}} - Y_{t_i}),$$

whenever the limit exists in probability. For Brownian motions $W, B$ with instantaneous correlation $\rho$,

$$\boxed{(\mathrm{d}W_t)^2 = \mathrm{d}t, \quad \mathrm{d}W_t \, \mathrm{d}B_t = \rho \, \mathrm{d}t, \quad \mathrm{d}t \, \mathrm{d}W_t = 0, \quad (\mathrm{d}t)^2 = 0.}$$

These are mnemonics for quadratic-variation limits, not ordinary algebraic infinitesimals. If

$$\mathrm{d}X_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}W_t + c_t \, \mathrm{d}B_t, \quad \mathrm{d}Y_t = p_t \, \mathrm{d}t + q_t \, \mathrm{d}W_t + r_t \, \mathrm{d}B_t,$$

then

$$\boxed{\mathrm{d}[X, Y]_t = \big(b_t q_t + c_t r_t + \rho b_t r_t + \rho c_t q_t\big) \, \mathrm{d}t.}$$

Only diffusion coefficients enter; finite-variation drifts contribute no quadratic covariation.

For the Itô process $\mathrm{d}X_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}W_t$,

$$\boxed{\mathrm{d}[X]_t = b_t^2 \, \mathrm{d}t, \qquad [X]_t = \int_0^t b_s^2 \, \mathrm{d}s.}$$

## 4.2 Itô's formula and the generator

**Theorem 4.1** (One-dimensional Itô formula)**.** *Let* $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ *and* $f \in C^{1,2}$*. Then*

$$\boxed{\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)(t,X_t)\,\mathrm{d}t + b_t f_x(t,X_t)\,\mathrm{d}W_t.}$$

*Equivalently,* $\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \frac{1}{2}f_{xx}\,\mathrm{d}[X]$.

For a Markov diffusion $\mathrm{d}X_t = b(t,X_t)\,\mathrm{d}t + a(t,X_t)\,\mathrm{d}W_t$, define

$$\mathcal{L}f = bf_x + \frac{1}{2}a^2 f_{xx}.$$

Then the transformed drift is $f_t + \mathcal{L}f$. Hence

$$\boxed{f_t + \mathcal{L}f = 0}$$

is the equation that makes $f(t,X_t)$ a local martingale. A separate boundedness or integrability argument is required to obtain a true martingale and use conditional expectations.

**Reliable application method.** Write the state variables as placeholders, compute all first derivatives and all second derivatives in noisy directions, substitute the processes, then group separately the $\mathrm{d}t$ and Brownian terms. Always check the domain of $f$, especially for $\log x$, $1/x$, and payoff kinks.

## 4.3 Reusable transformation rules

If $X_t > 0$ and $\mathrm{d}X_t/X_t = \alpha_t\,\mathrm{d}t + \beta_t\,\mathrm{d}W_t$, then

$$\boxed{\mathrm{d}\log X_t = \left(\alpha_t - \frac{1}{2}\beta_t^2\right)\mathrm{d}t + \beta_t\,\mathrm{d}W_t.}$$

The term $-\beta_t^2/2$ is the volatility correction. If $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ and $X_t \neq 0$, then

$$\boxed{\mathrm{d}\left(\frac{1}{X_t}\right) = \left(-\frac{a_t}{X_t^2} + \frac{b_t^2}{X_t^3}\right)\mathrm{d}t - \frac{b_t}{X_t^2}\,\mathrm{d}W_t.}$$

For continuous Itô processes,

$$\boxed{\mathrm{d}(X_tY_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}[X,Y]_t.}$$

For $f \in C^{1,2}([0,T]\times\mathbb{R}^2)$,

$$\boxed{\begin{aligned}\mathrm{d}f(t,X_t,Y_t) = {} & f_t\,\mathrm{d}t + f_x\,\mathrm{d}X_t + f_y\,\mathrm{d}Y_t \\ & + \frac{1}{2}f_{xx}\,\mathrm{d}[X]_t + f_{xy}\,\mathrm{d}[X,Y]_t + \frac{1}{2}f_{yy}\,\mathrm{d}[Y]_t.\end{aligned}}$$

The mixed derivative has no factor $1/2$.

### 4.4 Canonical SDEs to solve and reuse

**Geometric Brownian motion.** For

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_0 > 0,$$

apply Itô to $\log S_t$:

$$\boxed{S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}.}$$

Thus $S_t > 0$, $\log S_t$ is Gaussian, and

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \frac{\mathrm{d}(S_t^p)}{S_t^p} = \left[p\mu + \frac{1}{2}p(p-1)\sigma^2\right]\mathrm{d}t + p\sigma\,\mathrm{d}W_t.$$

**Ornstein–Uhlenbeck process.** For

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0,$$

the integrating factor $e^{\kappa t}$ yields, for $s \le t$,

$$\boxed{X_t = m + (X_s - m)e^{-\kappa(t-s)} + \sigma\int_s^t e^{-\kappa(t-u)}\,\mathrm{d}W_u.}$$

Therefore

$$\boxed{\mathbb{E}[X_t \mid \mathcal{F}_s] = m + (X_s - m)e^{-\kappa(t-s)}, \quad \mathrm{Var}(X_t \mid \mathcal{F}_s) = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(t-s)}\big).}$$

**Exponential martingale and Brownian moments.** Two particularly useful drift cancellations are

$$\boxed{\exp\left(\lambda W_t - \frac{1}{2}\lambda^2 t\right) \text{ is a martingale}, \qquad W_t^2 - t \text{ is a martingale.}}$$

More generally, applying Itô's formula to $W_t^n$ gives the recursion

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[W_t^n] = \frac{n(n-1)}{2}\mathbb{E}[W_t^{n-2}],$$

whenever the stochastic integral is integrable.

### 4.5 Several factors and instantaneous hedging

If the unhedged shock is $\sigma_X\,\mathrm{d}W_t - h\sigma_Y\,\mathrm{d}B_t$, with $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$, its variance rate is

$$q(h) = \sigma_X^2 + h^2\sigma_Y^2 - 2h\rho\sigma_X\sigma_Y.$$

The minimum-variance hedge is

$$\boxed{h^* = \rho\frac{\sigma_X}{\sigma_Y}, \qquad q(h^*) = \sigma_X^2(1 - \rho^2).}$$

Only the component correlated with the hedging asset can be removed.

**Lecture 4 checklist**

You should be able to:

- translate an Itô process between differential and integral forms;
- identify the time and state variables and write every driving SDE;
- compute $f_t$, $f_x$, and $f_{xx}$ before applying Itô's formula;
- calculate covariations, including $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$ for correlated drivers;
- retain all terms of order $\mathrm{d}t$ and separate the final drift and diffusion terms;
- use product, quotient, and integration-factor formulas for Itô processes;
- solve standard SDEs such as GBM and Ornstein–Uhlenbeck processes;
- check the domain, $C^{1,2}$ regularity, and integrability conditions;
- state the probability measure and financing convention in a financial application.

# 5 Lecture 5 — Self-Financing, Black–Scholes PDE, and Greeks

**Summary.** Self-financing is a cash-flow constraint, not merely a portfolio-value identity. In a complete Brownian market, a claim can be replicated and no arbitrage makes its price unique. Matching the diffusion of a self-financing hedge yields delta and matching the drift yields the Black–Scholes PDE. The physical drift cancels from the replication argument; the PDE solution gives the call and put formulas, while the Greeks describe hedge and parameter sensitivities. A continuous dividend yield changes carry from $r$ to $r - q$.

## 5.1 Market model, strategies, and self-financing

The basic market contains a stock and a bank account:

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t, \qquad B_t = e^{rt}.$$

The standard model assumes continuous GBM dynamics and constant parameters. The replication argument additionally uses frictionless predictable trading, common borrowing and lending rate, admissible strategies, and no arbitrage. These model and market assumptions play different logical roles.

**Definition 5.1** (Trading strategy and portfolio value)**.** A predictable strategy consists of stock and bank-account holdings $(\Delta_t, \beta_t)$. Its marked-to-market value is

$$\boxed{X_t = \Delta_t S_t + \beta_t B_t.}$$

**Definition 5.2** (Self-financing)**.** With no dividends, a strategy is self-financing when

$$\boxed{X_t = X_0 + \int_0^t \Delta_u\,\mathrm{d}S_u + \int_0^t \beta_u\,\mathrm{d}B_u,}$$

or $\mathrm{d}X_t = \Delta_t\,\mathrm{d}S_t + \beta_t\,\mathrm{d}B_t$. Equivalently,

$$\boxed{\mathrm{d}X_t = \Delta_t\,\mathrm{d}S_t + r(X_t - \Delta_t S_t)\,\mathrm{d}t.}$$

The value identity does not imply the gain identity. Rebalancing changes the composition of a self-financing portfolio, not its current wealth. In discrete time, this constraint is

$$S_{t_n}(\Delta_n - \Delta_{n-1}) + B_{t_n}(\beta_n - \beta_{n-1}) = 0.$$

If the stock pays a continuous dividend yield $q$, a stock position earns both price gains and dividends. The self-financing equation becomes

$$\boxed{\mathrm{d}X_t = \Delta_t(\,\mathrm{d}S_t + qS_t\,\mathrm{d}t) + \beta_t\,\mathrm{d}B_t.}$$

## 5.2 Replication and the Black–Scholes PDE

Let a European claim have payoff $H = g(S_T)$, and suppose its value is $V_t = p(t, S_t)$, with $p(T, s) = g(s)$. Itô's formula gives

$$\mathrm{d}V_t = \left(p_t + \mu S_t p_s + \frac{1}{2}\sigma^2 S_t^2 p_{ss}\right)\,\mathrm{d}t + \sigma S_t p_s\,\mathrm{d}W_t.$$

Match this with a self-financing portfolio. Equality of the diffusion terms first gives the hedge

$$\boxed{\Delta_t = p_s(t, S_t).}$$

Equality of the drift terms then gives the PDE.

**Theorem 5.3** (Black–Scholes PDE with dividend yield)**.** *Under the replication and no-arbitrage assumptions,*

$$\boxed{\begin{cases} p_t + (r-q)sp_s + \dfrac{1}{2}\sigma^2 s^2 p_{ss} - rp = 0, \quad t < T,\ s > 0, \\ p(T, s) = g(s). \end{cases}}$$

*The replicating holdings are*

$$\boxed{\Delta_t = p_s(t, S_t), \qquad \beta_t = \frac{p(t, S_t) - S_t p_s(t, S_t)}{B_t}.}$$

*For $q = 0$, this is the standard Black–Scholes PDE.*

The physical drift $\mu$ cancels because the claim and hedge have identical instantaneous stock exposure. Volatility remains through quadratic variation, and $r - q$ is the net carry of holding the stock.

**Completeness and no arbitrage.** Completeness means that traded assets span all modeled Brownian shocks; it allows the diffusion term to be replicated. No arbitrage then forces a claim and its replicating strategy to have the same price. With extra unspanned shocks or jumps, the market is generally incomplete and the same matching argument may fail.

## 5.3 Solving the PDE without a change of measure

Let $\tau = T - t$. Successively reverse time, remove discounting, set $x = \log s$, and remove the remaining transport term. This transforms the Black–Scholes PDE into a heat equation. Equivalently, its Gaussian-kernel solution is

$$\boxed{p(t, s) = e^{-r\tau}\mathbb{E}\left[g\left(s\exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}Z\right\}\right)\right], \quad Z \sim \mathcal{N}(0, 1).}$$

Here this formula is used as the explicit solution of the PDE; no change-of-measure argument is required in Lecture 5.

### 5.4 European call, put, and put–call parity

For $\tau = T - t$, define

$$\boxed{d_1 = \frac{\log(s/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.}$$

Then

$$\boxed{\begin{aligned} C(t,s) &= se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \\ P(t,s) &= Ke^{-r\tau}\Phi(-d_2) - se^{-q\tau}\Phi(-d_1). \end{aligned}}$$

The corresponding parity relation is

$$\boxed{C(t,s) - P(t,s) = se^{-q\tau} - Ke^{-r\tau}.}$$

For $q = 0$, the factor $e^{-q\tau}$ disappears. Useful consistency checks are

$$(se^{-q\tau} - Ke^{-r\tau})^+ \le C(t,s) \le se^{-q\tau},$$

the terminal limit $C(T,s) = (s-K)^+$, and the identity

$$\boxed{se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2),}$$

where $\varphi$ is the standard normal density.

### 5.5 Greeks to know and interpret

For the call,

$$\boxed{\begin{aligned} \Delta_C &= e^{-q\tau}\Phi(d_1), \\ \Gamma_C &= \frac{e^{-q\tau}\varphi(d_1)}{s\sigma\sqrt{\tau}}, \\ \mathrm{Vega}_C &= se^{-q\tau}\varphi(d_1)\sqrt{\tau}, \\ \Theta_C &= -\frac{se^{-q\tau}\sigma\varphi(d_1)}{2\sqrt{\tau}} + qse^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2), \\ \mathrm{Rho}_C &= K\tau e^{-r\tau}\Phi(d_2). \end{aligned}}$$

Here theta means calendar-time sensitivity $\partial C/\partial t$. Put–call parity gives

$$\boxed{\Delta_P = e^{-q\tau}(\Phi(d_1) - 1), \quad \Gamma_P = \Gamma_C, \quad \mathrm{Vega}_P = \mathrm{Vega}_C, \quad \mathrm{Rho}_P = -K\tau e^{-r\tau}\Phi(-d_2).}$$

Delta is the number of shares in the replicating hedge. Gamma measures how quickly delta changes; it is therefore a direct indicator of rebalancing risk. Vega measures sensitivity to the model volatility, theta to the passage of calendar time, and rho to the interest rate. Greeks are local derivatives, not bounds on finite losses.

### 5.6 Verification and discrete hedging

If a smooth $p$ solves the PDE and its stochastic integral is a true martingale, applying Itô's formula verifies both descriptions:

$$\boxed{\mathrm{d}p(t,S_t) = p_s(t,S_t)(\,\mathrm{d}S_t + qS_t\,\mathrm{d}t) + r\big(p(t,S_t) - S_t p_s(t,S_t)\big)\,\mathrm{d}t.}$$

Thus $\Delta = p_s$, together with the cash holding above, is self-financing and reaches $g(S_T)$.

In practice, delta is frozen over each interval of a trading grid. The leading price-move term not removed by a delta-only hedge is

$$\boxed{\frac{1}{2}\Gamma(\Delta S)^2.}$$

Discrete trading, transaction costs, jumps, stochastic or misspecified volatility, dividends, funding differences, and omitted state variables all create hedging errors outside the ideal model.

### Lecture 5 checklist

- state the stock and bank-account dynamics, distinguish model from market assumptions, and specify the payoff and terminal condition;
- define the portfolio holdings and value, then impose predictability and the self-financing condition;
- apply Itô's formula to a candidate option value and match diffusion and drift terms to identify delta and derive the Black–Scholes PDE;
- state the Black–Scholes call and put formulas, use put–call parity, incorporate a continuous dividend yield, and interpret delta, gamma, theta, vega, and rho;
- check regularity, boundary behavior, integrability, self-financing, and terminal replication, without invoking a change of measure.